

WEEK 7 – POWER BI REPORT

STUDENT PERFORMANCE – DATA CLEANING & TRANSFORMATION

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Tool Used: Microsoft Power BI

Project: Student Performance Analytics

1. Introduction

The Week 7 Power BI project focuses on cleaning, transforming, and preparing the **Student Performance in Exams** dataset for analysis. The dataset contains student information such as gender, race/ethnicity, parental education, lunch type, test preparation course, and scores in Mathematics, Reading, and Writing.

The main objective is to prepare a structured and reliable dataset that can be used to create an interactive Student Performance Dashboard in Week 8.

2. Objectives

The objectives of this project are:

- To import the Student Performance dataset into Power BI.
 - To inspect and understand the dataset.
 - To identify missing values, errors, and duplicate records.
 - To clean and standardize the data.
 - To create calculated columns such as Total Score and Average Score.
 - To classify students based on performance.
 - To create Pass/Fail classification.
 - To unpivot subject-wise score data.
 - To prepare the data model.
 - To create basic DAX measures.
 - To validate the transformed dataset.
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3. Dataset

The project uses the **Student Performance in Exams** dataset.

Important Attributes

- Gender
- Race/Ethnicity
- Parental Education
- Lunch
- Test Preparation Course
- Math Score
- Reading Score
- Writing Score

These attributes help analyze student academic performance and identify differences between different student groups.

4. Data Import

The dataset was imported into **Power BI Desktop** using:

Get Data → Excel

After importing the dataset, **Transform Data** was selected to open the **Power Query Editor** for data cleaning and transformation.

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor'S Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master'S Degree	Standard	None
4	Male	Group A	Associate'S Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate'S Degree	Standard	None
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate'S Degree	Standard	None
12	Male	Group D	Associate'S Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master'S Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master'S Degree	Free/Reduced	Completed
20	Female	Group C	Associate'S Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed

5. Data Inspection

The dataset was inspected in Power Query Editor to understand its structure.

The following checks were performed:

- Reviewed available rows and columns.
- Identified numerical and categorical attributes.
- Checked the data types.
- Identified missing values.
- Checked for errors.
- Checked for duplicate records.

This step helped ensure that the dataset was suitable for further analysis.

6. Data Cleaning

The following data cleaning operations were performed:

- Removed duplicate records where appropriate.
- Handled missing or invalid values.
- Corrected incorrect data types.
- Standardized inconsistent categorical values.
- Renamed columns where required for better understanding.

These steps improved the quality and consistency of the dataset.

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master's Degree	Standard	None
4	Male	Group A	Associate's Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate's Degree	Standard	None
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate's Degree	Standard	None
12	Male	Group D	Associate's Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master's Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master's Degree	Free/Reduced	Completed
20	Female	Group C	Associate's Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed

Query Settings

PROPERTIES

Name: StudentsPerformance

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Capitalized Each Word
- Changed Type1
- Cleaned Text
- Trimmed Text
- Added Index
- Renamed Columns
- Reordered Columns
- Added Custom
- Added Custom1
- Changed Type2
- Added Conditional Column

7. Categorical Data Transformation

The following categorical attributes were reviewed and standardized:

- Gender
- Race/Ethnicity
- Parental Education
- Lunch

- Test Preparation Course

The category names and values were standardized to maintain consistency during analysis.

8. Creating Calculated Columns

Calculated columns were created in Power Query to derive useful performance information.

Total Score

The Total Score was calculated using:

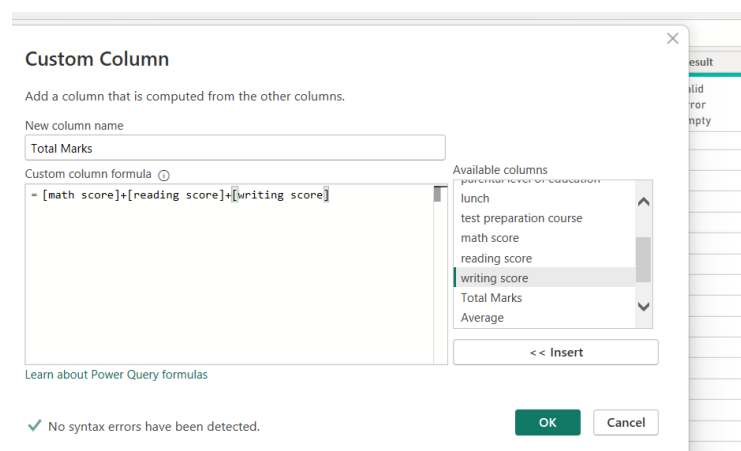
Total Score = Math Score + Reading Score + Writing Score

Average Score

The Average Score was calculated using:

Average Score = Total Score / 3

These columns provide an overall measure of each student's academic performance.



Custom Column

Add a column that is computed from the other columns.

New column name

Average

Custom column formula

= [Total Marks]/3

Available columns

parent level of education

lunch

test preparation course

math score

reading score

writing score

Total Marks

Average

< < Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

9. Performance Grade

A performance grade was created using conditional logic.

Grade	Performance
A	High Performance
B	Good Performance
C	Average Performance
F	Low Performance

This classification makes it easier to analyze students according to their overall performance.

Table.AddColumn(#"Added Conditional Column", "Result", each if [#"Average "] >= 40 then "Pass" else "Fail")

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Grade

Column Name

Average

Operator

is greater than or...

Value

80

Output

A

If

Then

Else If

Then

Else If

Then

Add Clause

Else

Fail

OK

Cancel

10. Pass/Fail Classification

A suitable pass criterion was defined and a **Conditional Column** was created.

The students were classified into:

- Pass
- Fail

This classification helps in calculating and analyzing the overall pass percentage.

The screenshot shows a software interface for creating a conditional column. At the top, a header bar displays a table structure with columns: 'TABLE', 'AVERAGE', 'NEW COLUMN', 'RESULT', 'FAIL', 'PASS', 'FAIL'. Below this, a dialog box titled 'Add Conditional Column' is open. It contains the instruction: 'Add a conditional column that is computed from the other columns or values.' The 'New column name' field is set to 'Result'. The main rule is defined in a table-like structure:

	Column Name	Operator	Value		Output
If	Average	is greater than or...	40	Then	Pass

Below the main rule, there is an 'Add Clause' button and an 'Else' section with a dropdown menu showing 'ABC 123' and a text field containing 'Fail'. At the bottom right, there are 'OK' and 'Cancel' buttons.

11. Unpivoting Subject Data

The subject score columns:

- Math Score
- Reading Score
- Writing Score

were selected and transformed using **Unpivot Columns**.

The resulting columns were renamed as:

- **Subject**
- **Score**

This transformation makes it easier to compare student performance across different subjects.

Queries [4] < ✓ ✕

StudentsPerformance

Student Marks

Student Details

Measure

Table.RenameColumns(#Unpivoted Columns,{"Attribute", "Subject"}, {"Value", "Score"})

	Student ID	Subject	Score
	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%
1		1 math score	72
2		1 reading score	72
3		1 writing score	74
4		2 math score	69
5		2 reading score	90
6		2 writing score	88
7		3 math score	90
8		3 reading score	95
9		3 writing score	93
10		4 math score	47
11		4 reading score	57
12		4 writing score	44
13		5 math score	76
14		5 reading score	78
15		5 writing score	75
16		6 math score	71
17		6 reading score	83
18		6 writing score	78
19		7 math score	88
20		7 reading score	95
21		7 writing score	92
22		8 math score	40
23		8 reading score	43
24		8 writing score	39

12. Data Model Preparation

After transformation, the dataset was prepared for the Power BI data model.

The following activities were performed:

- Reviewed the transformed table.
- Removed unnecessary columns.
- Ensured appropriate keys and relationships.
- Created additional tables where required.

This prepared the dataset for analysis and visualization.

13. Loading the Data

After completing the transformations, **Close & Apply** was selected to load the data into Power BI.

The transformed columns and unpivoted subject data were then verified in Power BI.

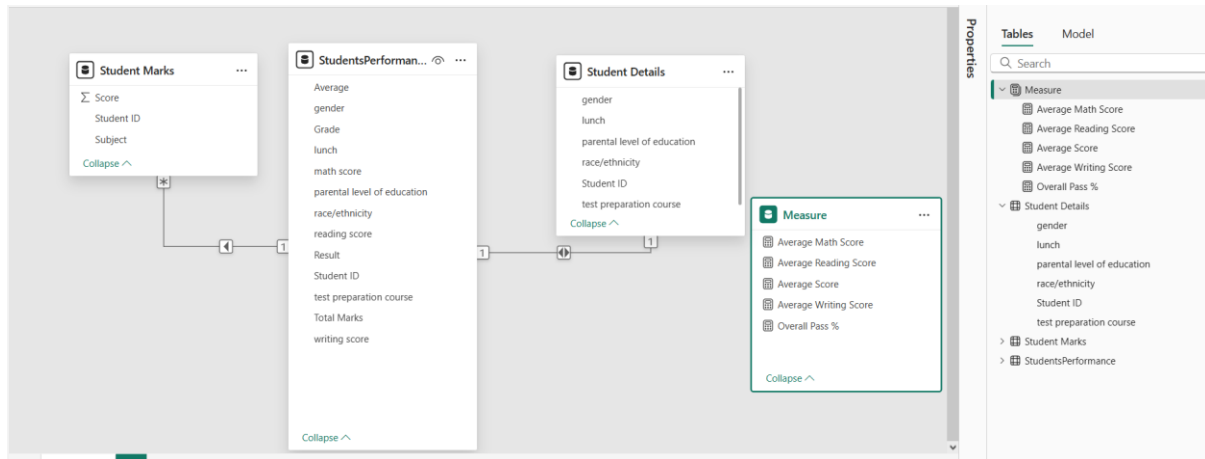
14. DAX Measures

Basic DAX measures were created to support performance analysis.

The measures include:

- Average Math Score
- Average Reading Score
- Average Writing Score
- Overall Average Score
- Overall Pass %

These measures will be used for creating KPIs and visualizations in the Week 8 dashboard.



15. Validation

The transformed dataset was validated by:

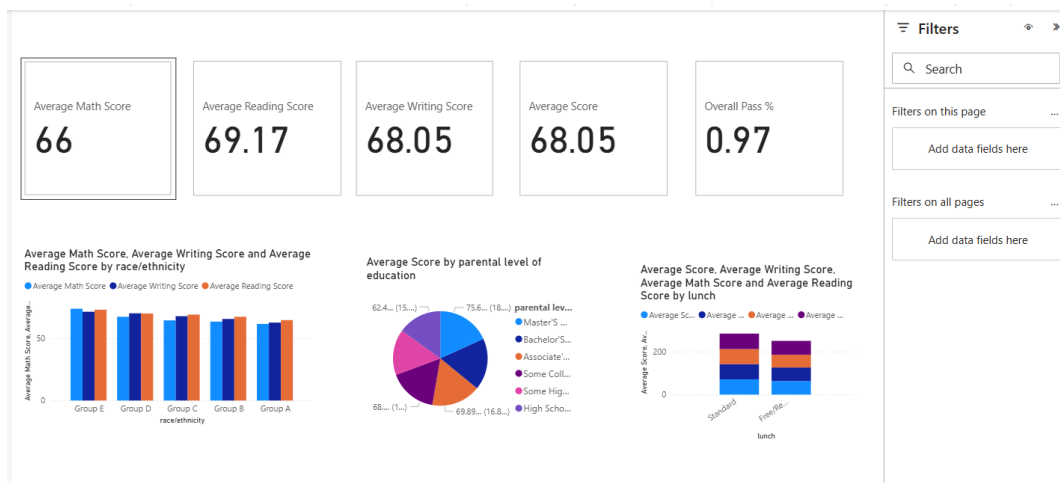
- Comparing Power Query results with the original dataset.
- Checking calculated scores.
- Verifying Pass/Fail classification.
- Verifying Grade classification.
- Confirming that the data model was ready for visualization.

16. Week 7 Output

The final output of Week 7 is a **cleaned, transformed, and structured Student Performance dataset** containing:

- Calculated attributes
- Performance grades
- Pass/Fail classification
- Unpivoted subject data
- Basic DAX measures

This prepared dataset will be used to develop the **Student Performance Analytics Dashboard in Week 8**.



17. Conclusion

The Week 7 Power BI project successfully prepared the Student Performance dataset for further analysis. Data cleaning, transformation, calculated columns, Pass/Fail classification, unpivoting, data modelling, and DAX measures were completed.

The cleaned dataset provides a strong foundation for developing an interactive dashboard in Week 8, where student performance can be analyzed using KPIs, charts, slicers, and other Power BI features.

Week 7 Workflow

Import → Inspect → Clean → Transform → Add Columns → Unpivot → Model → DAX → Validate

